HEIG-VD HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

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Formal verification
Verification of a math computer
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Module to verify

We want to verify a module able to calculate $a + a + b - c$. The three operands are supplied at the input, and the result is produced at the output after some latency. Both interfaces use a valid/ready handshake : a transfer occurs on the active clock edge when both signals are asserted. At the input, the source asserts input-valid and the design asserts input-ready ; at the output, the design asserts output-valid and the receiver asserts output-ready.
The `math_computer` uses records for its input and output interfaces. The PSL files define aliases such as `valid_i` and `result_o` for record fields. These aliases shorten property expressions ; they refer to the same signals and do not change the design behavior.
The module has been built with two different architectures : datapath-control, and pipeline.
The assertions are split across three files. Start with the shared file for interface and functional properties, then use the architecture-specific files only for properties that depend on that implementation :

- `math_computer.psl` : Applied to both designs
- `math_computer_only_datapath.psl` : Applied only to the datapath-control version
- `math_computer_only_pipeline.psl` : Applied only to the pipeline version

For the pipeline, consider checking that results correspond, in order, to computations accepted at the input. Do not assume a particular latency unless the specification requires one. The FIFO examples in `cours/code/fifo_anyconst` may help with tracking accepted transactions.